

1. Evaluate the following limits using l'Hospital's rule where appropriate.

(a) $\lim_{x \rightarrow 1} \frac{x^4 - 4x^3 - 30x^2 + 68x - 35}{3x^3 + 18x^2 - 45x + 24}$

(b) $\lim_{x \rightarrow -\infty} \frac{e^{2x} + 6x^2}{x^2 - e^{5x}}$

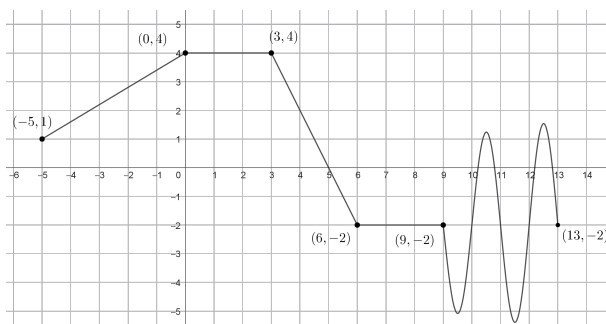
2. Determine whether each sequence converges or diverges. If the sequence converges, find its limit.

(a) $\left\{ \frac{7n+8}{3^n-5} \right\}$

(b) $\left\{ (-1)^n \frac{\sqrt{5+16n^2}}{\sqrt{8+25n^2}} \right\}$

3. BC Place in Vancouver is hosting some world cup games this year. If there are 45 rows of seats and the number of seats increases as they are further from the centre. The first row has 100 seats the second has 150, the third 200, and so on. How many seats are there?

4. Given $f(x)$ below calculate the following



(a) $\int_{-5}^0 f(x) dx$

(b) $\int_0^9 f(x) dx$

(c) Circle the largest:

$\int_0^9 f(x) dx, \quad \int_0^9 |f(x)| dx, \quad \int_0^{13} f(x) dx$

5. Use a Riemann sum with the right endpoint method and $n = 4$ to approximate the integral. (Round your answers to four decimals.)

$$\int_3^{11} \ln x dx$$

6. Use Riemann sums to evaluate the integral below. Provided that:

$$\sum_{k=1}^n c = cn, \quad \sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$\int_0^4 2 - x + 2x^2 dx$$

7. Evaluate each integral.

(a) $\int \frac{\sqrt{x} + (x+1)^2}{2\sqrt{x}} dx$

(b) $\int_0^6 |x-3| dx$

(c) $\int \frac{\sec^2(\ln x)}{x} dx$

(d) $\int (2x+3)^2 \cos x dx$

(e) $\int_0^3 \frac{4x^2}{(5+x^3)^2} dx$

8. Find the area enclosed by the curves of: $y = x^3 + 8$ and $y = 2x^2 + 4x$.

9. Evaluate the following improper integrals.

(a) $\int_0^\infty \frac{8e^{-x}}{1+e^{-x}} dx$

(b) $\int_2^6 \frac{5}{\sqrt{x-2}} dx$

10. The demand function for a product is $p = 144 - x^2$, and the supply function is $p = 48 + \frac{x^2}{2}$.

(a) Find the equilibrium point.

(b) Sketch a graph of the scenario indicating the equilibrium point, consumers' surplus and producers' surplus.

(c) Find the consumers' surplus.

11. Provided that the income distribution for Addie-Land is described by the function $f(x) = \frac{7x}{10} + \frac{3x^3}{10}$.

(a) Compute $f(0.5)$ and interpret your result.

(b) Find the coefficient of inequality (Gini coefficient) for Addie-Land.

(c) Given the Gini Coefficient for Canada is 0.311, which of the two countries has the more equitable wealth distribution?

12. Given $f''(x) = \frac{2}{\pi} + \sin(x)$, $f'(\pi/2) = 0$, $f(0) = 2$ find $f(x)$.

13. Solve the initial value problem for y : $\frac{dy}{dx} = (x^2 - x)y$ when $y(2) = 10$.

14. The rate of change of the population of Addie-Land is proportional to the square of time t (the years passed since confederation). At confederation there is 100 people and after 2 years there is 388 people.

(a) Write the differential equation and initial conditions to represent this situation.

(b) How many years after confederation until the population is over 1000?

15. Show that $f(x) = \frac{x}{\sqrt{x^2+9}}$ is **not** a probability density function on the interval $0 \leq x \leq 4$.

16. In Addie-Land, the temperature is normally distributed with a mean of 20°C and has a standard deviation of 8°C . Find the probability that a random day has a temperature between 5°C and 13°C .

17. Given that $g(x) = \frac{1}{24}(9 - x^2)$ is a probability density function for the random variable X on the interval $[-1, 2]$, find the following:

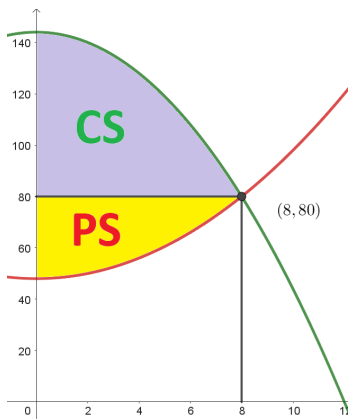
(a) $P(X > 1)$

(b) the mean of X

(c) the variance of X

Answers:

1. (a) $\frac{-4}{3}$
(b) 6
2. (a) 0
(b) $\frac{4}{5}$
3. 54000
4. (a) $\frac{25}{2}$
(b) 9
(c) $\int_0^9 |f(x)| dx$
5. 16.3009
6. $\frac{128}{3}$
7. (a) $\frac{1}{2}x + \frac{1}{5}x^{5/2} + \frac{2}{3}x^{3/2} + x^{1/2} + c$
(b) 9
(c) $\tan(\ln x) + c$
(d) $(2x + 3)^2 \sin x + (8x + 12) \cos x - 8 \sin x + c$
(e) $\frac{9}{40}$
8. $\frac{64}{3}$
9. (a) $8 \ln(2)$
(b) 20
10. (a) (8, 80)
(b)



(c) $\frac{1024}{3} \approx 341.33$

11. (a) 0.3875, the poorest half of the population have 38.75 % of the income.
(b) $\frac{3}{20} = 0.15$
(c) Addie-Land
12. $f(x) = \frac{1}{\pi}x^2 - \sin x - x + 2$
13. $y = 10e^{x^3/3 - x^2/2 - 2/3}$
14. (a) $\frac{dP}{dt} = kt^2, P(0) = 100, P(2) = 338$
(b) after 2.92 years
15. $\int_0^4 \frac{x}{\sqrt{x^2 + 9}} dx = 2 \neq 1$, therefore not a probability density function.
16. 0.1593
17. (a) $\frac{5}{18} \approx 0.2778$
(b) $\frac{13}{32} \approx 0.4063$
(c) $\frac{3507}{5120} \approx 0.6850$